BOUNDED BELOW COMPLEXES OF INJECTIVES COMPUTE EVERY RIGHT DERIVED FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7178-7194, following the statement at 7164-7176.

Editorial note. No omissions: the essentially-constant chain is displayed.

Allowed standard prerequisites (editorial).

- Lifting against injective resolutions. lemma-morphisms-lift (carded L26169, cited at ~7181). Gives the left inverse of any quasi-isomorphism out of I.

Source reference. Original definition 05SX: sources/stacks-derived/pinned-original-source.tex, source lines 5207–5207. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 0157: sources/stacks-derived/pinned-original-source.tex, source lines 5495–5495. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 05S9: sources/stacks-derived/pinned-original-source.tex, source lines 4753–4753. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 013P: sources/stacks-derived/pinned-original-source.tex, source lines 6603–6603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $I \in \mathrm{Ob}(\mathcal{A})$ be an injective object. Let $I^\bullet$ be a bounded below complex of injectives in $\mathcal{A}$.*

1. *$I^\bullet$ computes RF relative to $Qis^+(\mathcal{A})$ for any exact functor $F : K^+(\mathcal{A}) \to \mathcal{D}$ into any triangulated category $\mathcal{D}$.*
2. *I is right acyclic for any additive functor $F : \mathcal{A} \to \mathcal{B}$ into any abelian category $\mathcal{B}$.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Part (2) is a direct consequences of part (1) and Definition 0157. To prove (1) let $\alpha : I^\bullet \to K^\bullet$ be a quasi-isomorphism into a complex. By Lemma 013P we see that α has a left inverse. Hence the category $I^\bullet/\mathrm{Qis}^+(\mathcal{A})$ is essentially constant with value $\mathrm{id} : I^\bullet \to I^\bullet$. Thus also the ind-object

$$I^\bullet/\mathrm{Qis}^+(\mathcal{A}) \longrightarrow \mathcal{D}, \quad (I^\bullet \to K^\bullet) \longmapsto F(K^\bullet)$$

is essentially constant with value $F(I^\bullet)$. This proves (1), see Definitions 05S9 and 05SX. □